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Invited Commentary | Health Informatics

Large Language Models—Misdiagnosing Diagnostic Excellence?

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When the results of the Goh et al study[1] were presented at a recent National Academies of Medicine meeting, the audience was amazed—and concerned. The randomized clinical trial assessed diagnostic performance by generalist physicians, who were asked to provide diagnoses for 6 simulated cases using either conventional online resources or a large language model (LLM) (ChatGPT Plus [GPT-4]; OpenAI) in addition to standard resources. The study also evaluated the ability of the LLM to solve the cases alone. The authors developed a rubric for measuring diagnostic performance in which blinded experts evaluated participants' overall clinical reasoning process, including their proposed final diagnosis, their differential diagnosis, and factors supporting or opposing the diagnoses on their list.

The study's principal finding was that physicians who had access to the LLM scored no better than the group who only used conventional resources. But the result that prompted consternation was the performance of the LLM alone, which scored significantly higher than either group of physicians. On hearing these results, more than one audience member wondered aloud, "Are we going to be out of a job?"

The Goh et al trial[1] is an important advance in the study of generative artificial intelligence (AI) for diagnosis. By examining how clinicians use GPT-4 without specific training in use of the LLM, the study provides a realistic assessment of how clinicians use these tools in actual practice—both now and for the foreseeable future. Measuring the quality of the diagnostic process, rather than simply evaluating the accuracy of their final diagnosis, is a nuanced approach that provides a more accurate assessment of diagnostic reasoning. Future studies of diagnostic reasoning should use this method. The study demonstrates that access to generative AI alone will not improve diagnostic outcomes; clinicians will require training to use these resources to their full potential.

The authors appropriately caution that the results of this study "should not be interpreted to indicate that LLMs should be used for diagnosis autonomously without physician oversight,"[1] but the finding that the LLM outperformed physicians will almost certainly be the headline finding. Since making diagnoses is central to clinician's professional identity, it is not surprising that the prospect of using LLM for diagnosis evokes both excitement and trepidation. As more studies are published demonstrating the diagnostic capabilities of LLMs, what does this mean for clinicians?

There are reasons to be skeptical that the performance of LLMs on simulated cases can generalize to the clinical practice setting environment. The study's cases were representative of common general practice diagnoses but are presented in an orderly fashion with the relevant history, physical examination, laboratory, and imaging results necessary to construct a prioritized differential diagnosis. Diagnosis in the clinical setting is an iterative—and complicated—process that takes place amid many competing demands and requires input from the patient, caregivers, and multiple clinicians in addition to objective data. Far from a linear process, diagnosis in the clinical practice setting involves progressively refining diagnoses based on new information, and the distinction between diagnosis and treatment is often blurred as clinicians incorporate treatment response into diagnostic reasoning.

How do LLMs perform at diagnosis under conditions closer to actual clinical practice? A recent study[2] evaluated the performance of LLMs on diagnosing and developing management plans for 4 common abdominal conditions, using a dataset consisting of anonymized real patient data. Information was presented to the LLM in a stepwise manner, and after each step, the LLM was asked to summarize the information and provide a diagnosis or request additional testing. Once the LLM provided a diagnosis, it was required to recommend a treatment plan. When confronted with this

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realistic clinical decision-making scenario, LLMs performed poorly: significantly worse than physicians for all but the simplest diagnoses. The LLMs also failed to consistently request appropriate diagnostic testing and frequently made incorrect treatment recommendations even after arriving at the correct diagnosis.

Continued refinement of LLMs may eventually address these limitations, and this may happen quickly given the rapidity with which the performance of LLMs has improved. But even if LLMs prove to be capable of iterative diagnosis based on evolving information, will they reduce harm from diagnostic errors? Here again, some skepticism is warranted.

As with all adverse events, missed and delayed diagnoses occur due to underlying system failures (latent errors) that allow errors made by individuals (active errors) to reach the patient and cause harm. The power of generative AI lies in its ability to help individual clinicians avoid cognitive errors[3]—failures of data gathering and synthesis—that are properly considered active errors. But cognitive errors do not occur in isolation from the system in which clinicians work; these errors may in fact be a symptom of working in a health care system rife with latent errors (inadequate staffing, production pressures, flawed electronic health records, and poor interprofessional communication among them) that increase the clinicians' cognitive load[4] and prevent them from making accurate decisions. Focusing on the generative potential of AI to prevent cognitive errors runs the risk of minimizing the role of systems failures in diagnostic error and drawing attention away from addressing these issues.

Achieving diagnostic excellence will require a system that potentiates clinicians' ability to make accurate and timely diagnoses and supports patients through their diagnostic and therapeutic journey. Generative AI will be a part of this system, but successful integration of LLMs into clinical diagnosis will require technical improvements, training of clinicians, and thoughtful integration of technology into the clinical environment. Even if these steps are achieved, clinicians and AI developers should temper their expectations; the vast potential of generative AI to improve diagnosis will not be achieved without corresponding improvement in clinicians' work environment. Those who view the results of the impressive study by Goh et al[1] as a threat to physicians' job security should be reassured. And if you do not believe me, the chatbot studied in the trial agrees:

The LLM was queried using the University of California, San Francisco, Versa secure generative AI platform on August 28, 2024. Versa was asked to respond in 100 words or less to the prompt "recent studies have shown that large language models can diagnose simulated cases with accuracy equal or superior to practicing physicians. Given these results, will LLMs eventually reach the point where they can replace doctors?"

The AI-generated response was as follows: "Large language models (LLMs) have shown promise in diagnosing simulated cases, but replacing doctors entirely is unlikely. LLMs can augment medical practice by providing decision support, processing large datasets, and offering diagnostic suggestions. However, the practice of medicine involves complex human interactions, ethical judgments, and physical examinations that cannot be fully replicated by AI. LLMs are tools that can enhance health care but not replace the nuanced skills and holistic approach provided by human physicians. The integration of LLMs should focus on collaboration, not replacement."

ARTICLE INFORMATION

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Additional Information: The UCSF Versa Chat application, which is UCSF's secure generative AI platform, was used in this commentary, and was prompted with a question relevant to the subject of this commentary. The prompt and Versa's response have been included verbatim in the commentary and have been clearly indicated as AI-generated content. The purpose of using AI was simply to illustrate one of the main points of this commentary.

REFERENCES

1. Goh E, Gallo R, Hom J, et al. Large language model influence on diagnostic reasoning: a randomized clinical trial. *JAMA Netw Open*. 2024;7(10):e2440969. doi:10.1001/jamanetworkopen.2024.40969

2. Hager P, Jungmann F, Holland R, et al. Evaluation and mitigation of the limitations of large language models in clinical decision-making. *Nat Med*. 2024. doi:10.1038/s41591-024-03097-1

3. Singh H, Giardina TD, Meyer AND, Forjuoh SN, Reis MD, Thomas EJ. Types and origins of diagnostic errors in primary care settings. *JAMA Intern Med*. 2013;173(6):418-425. doi:10.1001/jamainternmed.2013.2777

4. Knees M, Raffel KE, Kissler M, et al. Cognitive load theory and its impact on diagnostic accuracy. Agency for Healthcare Research and Quality; 2024. Publication No. 24-0010-3-EF.

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